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- **Background and Aims** Stomatal regulation allows plants to promptly respond to water stress. However, our understanding of the impact of above and belowground hydraulic traits on stomatal regulation remains incomplete. The objective of this study was to investigate how key plant hydraulic traits impact transpiration of maize during soil drying. We hypothesize that the stomatal response to soil drying is related to a loss in soil hydraulic conductivity at the root–soil interface, which in turn depends on plant hydraulic traits.
- **Methods** We investigate the response of 48 contrasting maize (*Zea mays*) genotypes to soil drying, utilizing a novel phenotyping facility. In this context, we measure the relationship between leaf water potential, soil water potential, soil water content and transpiration, as well as root, rhizosphere and aboveground plant traits.
- **Key Results** Genotypes differed in their responsiveness to soil drying. The critical soil water potential at which plants started decreasing transpiration was related to a combination of above and belowground traits: genotypes with a higher maximum transpiration and plant hydraulic conductance as well as a smaller root and rhizosphere system closed stomata at less negative soil water potentials.
- **Conclusions** Our results demonstrate the importance of belowground hydraulics for stomatal regulation and hence drought responsiveness during soil drying. Furthermore, this finding supports the hypothesis that stomata start to close when soil hydraulic conductivity drops at the root–soil interface.

Key words: Critical soil water potential, transpiration, leaf water potential, rhizosheath, rhizosphere, plant hydraulic conductance, root length, soil drying, *Zea mays*, phenotyping.

INTRODUCTION

In light of the ongoing drastic climate change, drought events are predicted to become more frequent globally, due to increased temperatures and changes in the hydrological cycle (IPCC 2022). Limited water supply is one of the largest impediments to food production worldwide. In combination with a growing world population, the pressure on available nutritional resources is increasing dramatically (Rivero *et al.*, 2022). Therefore, breeding programmes are aiming to increase crop yield resistance and resilience to environmental stress.

Stomatal regulation plays an important role for plant survival under drought and for performance in terms of yield as it allows plants to promptly react to water stress (Buckley, 2019; Deans

et al., 2020). Stomata regulate the water flow and gas exchange across the soil–plant–atmosphere continuum (SPAC). Although the coordination between stomatal closure and aboveground hydraulics has been extensively studied (Scoffoni *et al.*, 2014; Bartlett *et al.*, 2016; Anderegg *et al.*, 2017; Henry *et al.*, 2019), our understanding of the impact of belowground hydraulics on stomatal regulation remains incomplete.

Water flow across the SPAC is driven by transpiration, which creates a suction in the roots. This tension drives the water flow from the soil to the roots along a gradient of potentials that can be described as a system of hydraulic resistances in series. The flow is proportional to the difference in water potentials and the variable hydraulic conductance of the components of the SPAC. The transpiration rate is set by atmospheric conditions and dependent

on leaf area. On a short timescale, it is regulated by stomatal opening and closing (Brodribb *et al.*, 2019; Buckley, 2019).

Sperry and Love (2015) and Sperry *et al.* (2016) have suggested a hydraulic framework to describe the physical constraints on transpiration and therefore stomatal functioning. They postulated that transpiration increases linearly with decreasing leaf water potential until a certain point. At this point, the resistance in some part of the SPAC increases to an extent that leaf water potential drops non-linearly for only a slight increase in transpiration rate. Rather than emerging from a mechanistic hypothesis, the premise of such models of stomatal regulation is that stomatal closure avoids the excessive drop in leaf water potential by responding to non-linearities in the relationship between leaf water potential and transpiration. In the context of soil drying, this drop in conductance and therefore the introduction of non-linearities has been proposed to be located belowground, especially in the soil, leading to a feed-forward type of stomatal regulation (Carminati and Javaux 2020). Recent studies have confirmed that the loss in belowground hydraulic conductivity (soil, roots and/or soil–root interface) represents the primary driver of stomatal closure and gas exchange (Rodríguez-Domínguez and Brodribb, 2020; Abdalla *et al.*, 2021, 2022; Abdalla and Ahmed, 2021; Bourbia *et al.*, 2021; Cai *et al.*, 2021, 2022; Koehler *et al.*, 2022).

This is not surprising considering that the hydraulic conductivity of the soil drops by more than ten orders of magnitude with soil drying (Draye *et al.*, 2010). The drop in conductance is associated with an enhanced water depletion in close proximity to the roots and a relatively high water flux around the roots, caused by the radial nature of water flow into the roots (Gardner 1965). Hence, under soil drying, plants experience a large drop of water potential around their roots. Additionally, as the soil water potential decreases, the hydraulic conductivity of the root–soil interface (Zarebanadkouki *et al.*, 2016) and of the root decrease (Parent *et al.*, 2009; Hachez *et al.*, 2012), which exacerbates the effect of soil drying on root water uptake.

The objective of this study is to expand our understanding of the physical constraints to water flow in soils and plants across the soil–root interface and how these constraints impact the emergent behaviour of stomatal water use regulation and, hence, transpiration response in drying soil. We used the critical soil water potential at which plants started decreasing transpiration in response to soil drying as a variable indicating differences in water use regulation (Cai *et al.*, 2022). Our general hypothesis is that the critical soil water potential is related to above and belowground traits (e.g. maximum transpiration, plant hydraulic conductance and the root surface that is effective in water uptake) that determine the development of water potential gradients around the roots during soil drying, based on the soil–plant hydraulic model of Carminati and Javaux (2020). In the following, we briefly introduce the different traits and discuss their implication for the overall hypothesis.

THEORY

The model by Carminati and Javaux (2020) proposes that stomatal closure is triggered by non-linearities in the relationship between leaf water potential and transpiration. This indicates that stomata close when the drop in potential from the bulk soil to the root surface, $\Delta\Psi_{\text{soil}} = \Psi_{\text{soil}} - \Psi_{s-r}$ (where Ψ_{s-r} is

effective water potential at the root–soil interface, and Ψ_{soil} is the soil water potential of the bulk soil, far from the root surface), approaches the loss in water potential within the plant, $\Delta\Psi_{\text{plant}} = \Psi_{s-r} - \Psi_{\text{leaf}}$ (where Ψ_{leaf} is the leaf water potential), i.e. when $\Delta\Psi_{\text{soil}} \approx \Delta\Psi_{\text{leaf}}$. The critical soil water potential ($\Psi_{\text{soil crit.}}$ [hPa]) at which transpiration starts to decline corresponds to the point when $\Delta\Psi_{\text{soil}} \approx \Delta\Psi_{\text{leaf}}$. This condition depends on soil and plant hydraulic variables, which in turn are dependent on the expression of plant and rhizosphere traits, namely: maximum stomatal conductance (here: indicated by maximum transpiration, considering the fairly constant humidity and energy balance over the course of the experiment), plant hydraulic conductance and the root surface area effective in water uptake.

Maximum transpiration ($Tr_{\text{max.}}$ [g s^{-1}]) sets the water demand and determines how quickly soil water resources are potentially depleted (Gholipour *et al.*, 2013). The faster water flows into the roots, the more severely water potential and hydraulic conductance drop at the soil–root interface due to the radial nature of water flow into the roots (Gardner 1965) and to the non-linearity of soil water flow equations. $\Delta\Psi_{\text{soil}}$ increases non-linearly with $Tr_{\text{max.}}$ and, for constant $Tr_{\text{max.}}$, it increases with decreasing Ψ_{soil} , because the drier the soil the larger is $\Delta\Psi_{\text{soil}}$ needed to sustain $Tr_{\text{max.}}$. It follows that $\Psi_{\text{soil crit.}}$ becomes less negative for increasing $Tr_{\text{max.}}$.

The dissipation in water potential within the plant tissues ($\Delta\Psi_{\text{plant}}$) is inversely proportional to the plant hydraulic conductance (K_{plant} [$\text{cm}^3 \text{hPa}^{-1} \text{s}^{-1}$]). Hence, the higher is K_{plant} , the lower is $\Delta\Psi_{\text{plant}}$, and the less negative is $\Psi_{\text{soil crit.}}$ at which $\Delta\Psi_{\text{soil}} \approx \Delta\Psi_{\text{plant}}$. Consequently, for plants with a high hydraulic conductance, the soil will more quickly become the limiting factor for the overall flow in the SPAC. Plants with a high conductance are expected to be more sensitive in their transpiration response to an increase in water potential gradients around roots in drying soils (Cai *et al.*, 2022). Note that this is an implication of the hypothesis that stomatal closure is driven by a loss of soil–plant hydraulic conductance (Carminati and Javaux, 2020). Contrastingly, in well-watered conditions at high vapour pressure deficit (VPD), when the largest dissipation of water potentials along the soil–plant continuum occurs in the plant rather than in the soil, a low plant hydraulic conductance will cause a very negative leaf water potential.

Root surface area effective in water uptake (A_{root} [cm^2]) determines the water supply to the plant. A smaller root surface effective in water uptake requires a faster flow rate of water in close proximity to the roots and larger gradients in soil water potential to sustain the same transpiration rate (Abdalla *et al.*, 2022). Plants with small A_{root} are expected to have a less negative $\Psi_{\text{soil crit.}}$.

From these considerations, we expect that: (1) plants exhibiting a higher maximum transpiration will close stomata and limit transpiration at higher (less negative) critical soil water potentials (Fig. 1A); (2) plants with a higher plant hydraulic conductance will limit transpiration at higher soil water potentials (Fig. 1B); and (3) plants with a smaller effective root surface area will limit transpiration at higher soil water potentials (Fig. 1C). Therefore, we postulate that there is a positive correlation between $\Psi_{\text{soil crit.}}$ with the following combination of traits: $Tr_{\text{max.}} * K_{\text{plant}} / A_{\text{root}}$. Solving the model by Carminati and Javaux (2020) yields the expected behaviour (Fig. 1).

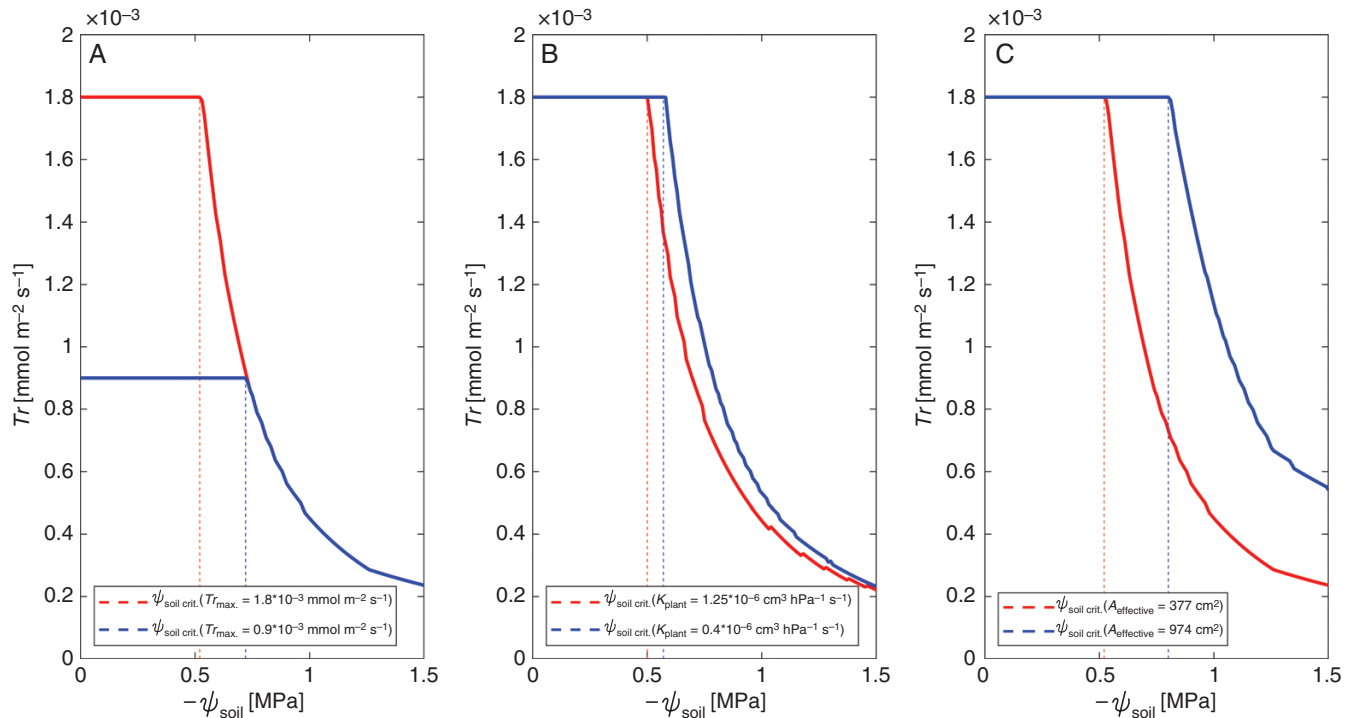


FIG. 1. Relationship between transpiration (Tr) and soil water potential (ψ_{soil}). Hypothesis: (A) plants with a higher maximum transpiration (Tr_{max} , red) will close stomata at a higher (less negative) critical soil water potential ($\psi_{\text{soil crit.}}$); (B) plants with a higher hydraulic conductance (K_{plant} , red) will close stomata at higher soil water potentials upon soil drying – note that this holds true for the case when the conductivity of plant tissues does not decline with water potential, which might not be the case for high VPD, low K_{plant} or plants with tissues extremely sensitive to a decline in water potential; and (C) plants with a lower root surface area ($A_{\text{effective}}$, red) effective in water uptake will close stomata at higher soil water potentials.

Finally, note that the proposed index $Tr_{\text{max}} \cdot K_{\text{plant}} / A_{\text{root}}$ is used to test if the general concept is valid, rather than to propose a new index or to reveal underlying physiological mechanisms of stomatal closure.

To test our hypotheses, we conducted a dry-down experiment with a selection of 48 maize (*Zea mays* L.) genotypes that were exposed to soil drying. During soil drying, we monitored soil moisture, transpiration and leaf water potential (both at pre-dawn and midday). Upon sampling, we quantified root dry biomass and the root adhering soil (rhizosheath; Ndour *et al.*, 2020).

MATERIALS AND METHODS

We performed a phenotyping experiment in which 48 maize (*Zea mays* L.) genotypes were screened for their response to soil drought and their expression of above and belowground traits. The experiment was carried out at the novel, fully automated ‘moving fields’-phenotyping facility of the Bavarian State Research Center for Agriculture (LfL) in Freising, Germany.

Plant material and growth

We used 48 maize genotypes (Supplementary Data Table S1) differing in drought responsiveness and plant traits. All genotypes originated from Germany, Austria or Switzerland. The

plants were grown under glasshouse conditions with a photoperiod of 14 h, with day/night air temperature averages of 20.8/18.9 °C and relative humidity averages of 52.1/56.1 % for 37 d, under optimal conditions. Glasshouse temperature and humidity were recorded and logged automatically (Pt1000, temperature and humidity sensor, RAM GmbH Mess- und Regeltechnik, Herrsching, Germany). Plants were grown in boxes (hereafter called mesocosms, Fig. 2B). Each mesocosm was filled with 10 kg of soil. One mesocosm consisted of two soil compartments, which were separated by a prism-shaped Styrofoam block (Fig. 2A). The compartments were hydraulically connected through a thin layer of soil above and below the Styrofoam (Fig. 2A). Each mesocosm contained two plants of the same genotype, one per compartment. Mineral fertilizer was applied three times [12 d after sowing (DAS), 18 DAS and 25 DAS] with 60 mL of the following substances dissolved in deionized water (1. run: 289 g KNO_3 , 114 g NH_4NO_3 , 92 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 48 L deionized water; 2. run: 433 g KNO_3 , 171 g NH_4NO_3 , 123 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 48 L deionized water; 3. run: 577 g KNO_3 , 229 g NH_4NO_3 , 215 g $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$ in 48 L deionized water).

We studied the variation in the response of maize genotypes to soil drought using a multifactorial design: the 48 maize genotypes were grown under two water treatments with four replicates each: well-watered and water-stressed. Plant drought stress was assessed by measuring the reduction in transpiration per genotype. Additionally, there were three control-mesocosms per treatment consisting of unplanted

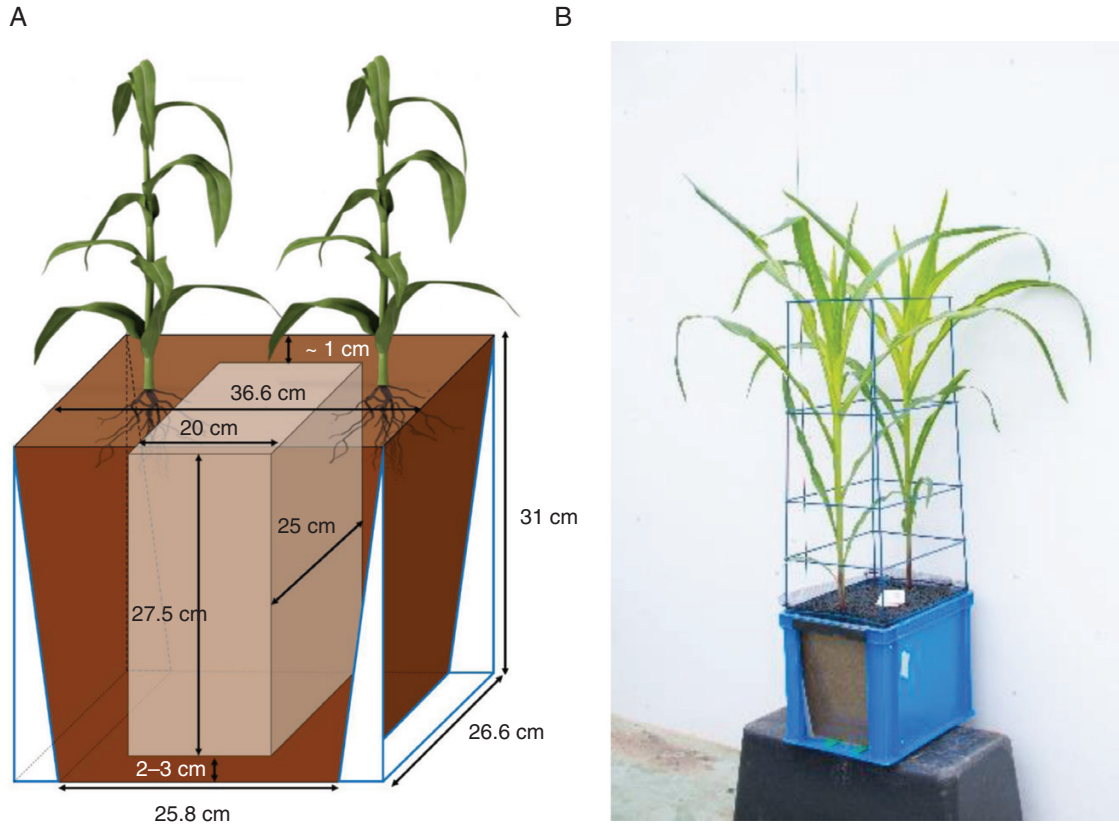


FIG. 2. Experimental set-up. Conceptual figure of the mesocosms, including dimensions (left). Each mesocosm consists of two compartments with one plant each (right, picture courtesy of N. Tyborski).

soil, resulting in 390 mesocosms ($48 \times 2 \times 4 + 6$) in total. The mesocosms were placed in completely random order and their distribution in the glasshouse was shuffled daily to minimize effects of differences in microclimate within the glasshouse.

Soil properties

The soil was excavated from an agricultural field [Freising, Germany, $48^{\circ}24'23.7''\text{N}$, $11^{\circ}41'24.4''\text{E}$, Ap-horizon of Haplic Luvisol (Humic, Loamic; WRB, 2015)], sieved through a 2-mm sieve to remove large plant residues, stones and coarse aggregates, and homogenized. To avoid soil layering and differences in bulk density between mesocosms, mesocosms were shaken automatically with a defined intensity and time while filling (17 s at 3000 r.p.m.).

The soil used in the present experiment has a slightly clayey loam texture (Supplementary Data Table S2). The hydraulic properties of the initial soil material were characterized by measuring the water retention and soil hydraulic conductivity curve using HYPROP2 (METER ENVIRONMENT, Munich, Germany) prior to the experiment. The hydraulic functions were fitted (Supplementary Data Table S3) using the Peters Durner Iden (PDI) parameterization (Peters, 2013, 2014; Iden and Durner, 2014).

Chemical properties of the initial soil material are provided in Supplementary Data Table S4.

Soil–plant hydraulic measurements

To minimize soil evaporation, the soil surface was covered with a mesh and a layer of plastic beads before application of the drought treatment. After a growing period of 37 d, when plants were optimally supplied with water, half of the mesocosms were exposed to water stress by withholding water while the other half remained constantly watered up to mean water potentials of -60 hPa. Mesocosms were weighed every morning at the same time. Daily transpiration was calculated as the difference in weight of each pot on successive days, plus the water added to the pots between two consecutive weighings [in the case of the well-watered treatment (Devi et al., 2010)]. Transpiration values were normalized in two steps to facilitate comparison between differently sized plants and to account for plant growth over time. First, a transpiration ratio (Tr_i) for each plant was calculated per day by dividing the transpiration of each drying pot (T_{dry}) by the average transpiration of the control pots of the same genotype in well-watered conditions (T_{ww}) (Kholová et al., 2010):

$$Tr_i = \frac{T_{dry}}{T_{ww}}$$

In a second step, Tr_i was normalized to obtain the normalized transpiration ratio (NTR_i) for each pot. This was done by calculating the initial average transpiration ratio for each pot (Tr_{av}) over the first 5 d after first withholding irrigation, when the plants were still in well-watered conditions. Daily Tr_i was divided by Tr_{av} as (Devi et al., 2010):

$$NTR_i = \frac{Tr_i}{Tr_{av}}$$

When the last drought-stressed plant reached an NTR of 0.5, meaning that the plants were stressed to the extent that they decreased transpiration by at least 50 %, the experiment was terminated. By that time, the most drought-stressed plant exhibited an NTR of 0.18.

Soil water content (θ) was measured periodically over the drying period using a time-domain reflectometer (TDR, Theta Probe type ML2X and HH2 moisture meter, Delta-T devices, Cambridge, UK). Soil water content was estimated for days without measurements by linearly fitting the soil water decrease over time to overall weight decrease replicate-wise. Subsequently, based on the obtained parameterization of that relationship, soil water contents were predicted from the weight (Fig. 3A). Soil water potential was measured utilizing soil water potential sensors (TEROS21, METER, Munich, Germany) in 56 of the mesocosms and estimated for the remaining 334 mesocosms from calculated soil water contents and from the water retention curve (Fig. 3B).

The critical soil water content at which plants started decreasing transpiration ($\theta_{crit.}$, Fig. 4 and Supplementary Data Fig. S1), an indicator for drought responsiveness, was determined by a linear plateau regression that was performed on the sigmoidal fit of the measured relationship between transpiration and soil water content (Ray and Sinclair, 1997). The critical soil water potential ($\Psi_{soil\ crit.}$) was calculated from $\theta_{crit.}$ using the soil water retention curve. The maximum transpiration ($Tr_{max.}$) was also obtained from the linear plateau regression. It was

defined as the y-axis intercept of the horizontal segment of the linear regression.

Leaf water potential (Ψ_{leaf}) was mostly determined at the fourth leaf stage (depending on availability) by day and under pre-dawn conditions by the use of the Scholander pressure chamber (Field plant water status console, UGT, Müncheberg, Germany). Under pre-dawn conditions, transpiration is assumed to be minimal and water potential in roots and soil are expected to be close to equilibrium (Ritchie and Hinckley, 1975). Thereby, the pre-dawn leaf water potential serves as a measure of soil water potential distribution, weighed according to the distribution of root conductance and including matrix as well as osmotic components of the soil water potential (Couvreur et al., 2012). The pre-dawn leaf water potential is an indicator of the soil water potential actually experienced by the plant, under the assumption that plant and soil had time to reach equilibrium. Due to the small size of the pots and the moderate level of drought, we supposed that this assumption was justified. During midday (1000–1100 h, mean glasshouse VPD = 1.41 kPa), leaves were collected and placed in air-sealed black bags that contained a moist piece of paper towel, to avoid a drop in leaf water potential (Knipfer et al., 2020; Rodriguez-Dominguez et al., 2022). Subsequently, Ψ_{leaf} was measured.

We inferred plant hydraulic conductance (K_{plant}) from the slope of the relationship between transpiration and leaf water potential of the well-watered plants. It is indeed generally assumed that plant hydraulic conductance sets the limits to water flow in wet soil (Passioura, 1980; Hopmans and Bristow, 2002):

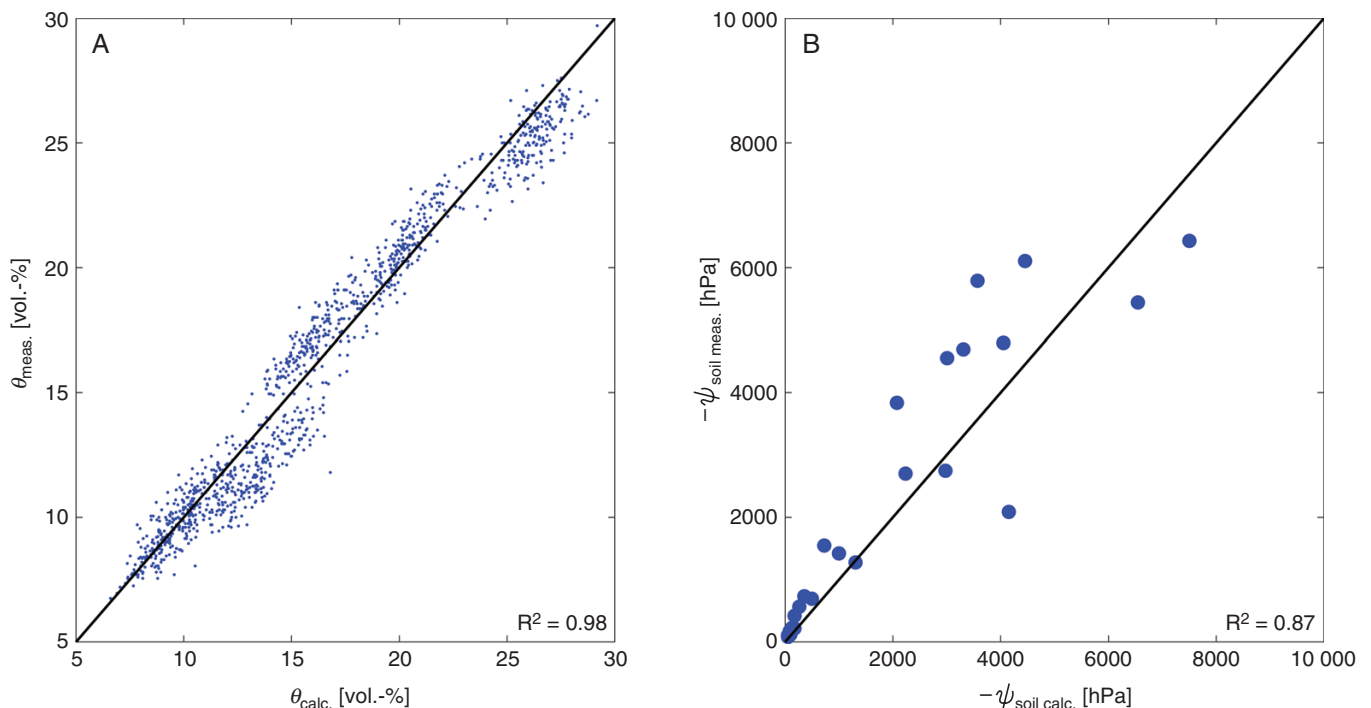


FIG. 3. (A) Relationship and fit between soil water content as measured by TDR ($\theta_{meas.}$) and soil water content as calculated ($\theta_{calc.}$) from the weight of the mesocosm; and (B) relationship between soil water potential as measured by the soil water potential sensors ($\Psi_{soil\ meas.}$) and soil water potential as calculated ($\Psi_{soil\ calc.}$) from calculated soil water contents via the soil water retention curve.

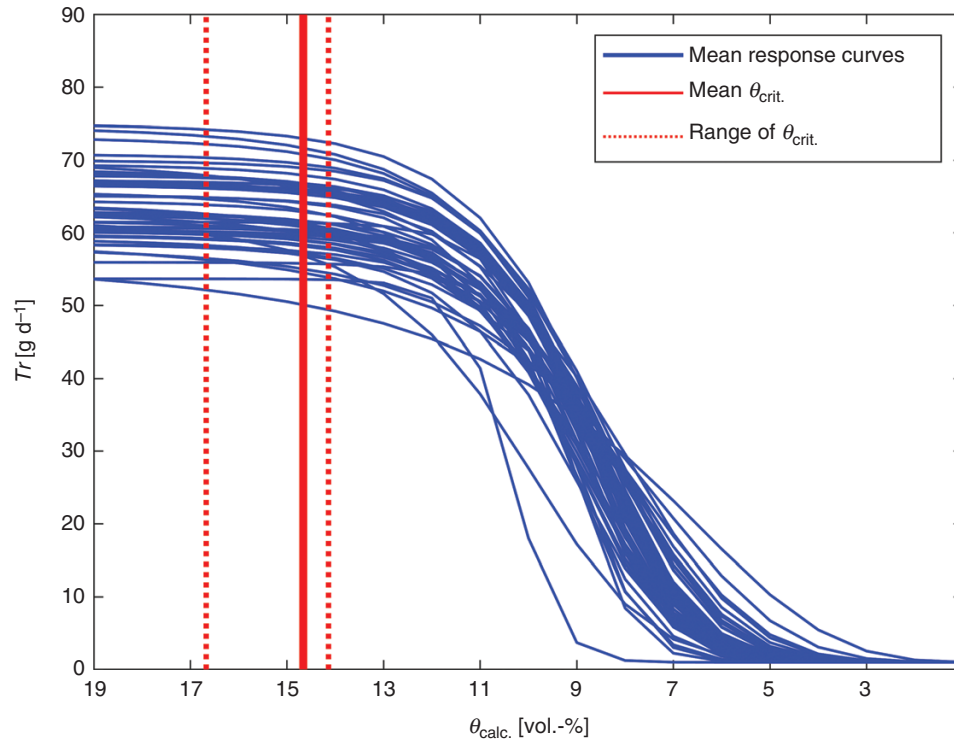


FIG. 4. Relationship between average transpiration (Tr) per genotype and the calculated soil water content ($\theta_{calc.}$), including the overall average critical soil water content ($\theta_{crit.}$) and the max. and min. ranges of $\theta_{crit.}$.

$$K_{plant} = \frac{Tr}{(\Psi_{soil} - \Psi_{leaf})}$$

where Tr is transpiration [g s^{-1}], Ψ_{leaf} [hPa] is the leaf water potential at that transpiration and Ψ_{soil} [hPa] is the soil water potential as experienced by the plant, which we obtained from the pre-dawn leaf water potential, Ψ_{PD} [hPa].

Sampling

After 27–37 d of soil drying, plants were harvested and above and belowground biomass were collected and quantified (Supplementary Data Note S1) for one of the two soil compartments. For the second soil compartment of the same mesocosm, roots were washed and scanned for a sub-selection of 38 genotypes (RhizoVision Explorer Version 2.0.2, Seethepalli *et al.*, 2020). Rhizosheath mass was determined by carefully extracting the roots from the soil and placing them into a self-made apparatus ('shakinator') which shook them with a defined intensity and time (Supplementary Data Fig. S2). The soil that adhered to the roots afterwards was defined as the rhizosheath and carefully stripped off by hand, dried at 105 °C and weighed (Supplementary Data Note S2; Ndour *et al.*, 2020).

In the following, root surface area effective in water uptake was assumed to be roughly related to the product of root biomass dry weight, as a measure of root length (Supplementary Data Fig. S3), and rhizosheath dry mass, as the rhizosheath can be seen as an extension of the effective root radius.

Statistics

All response variables (i.e. above and belowground traits, and emerging traits) were tested for normal distribution (Shapiro–Wilk test) and homogeneity of variance (Bartlett test). Based on this, either an independent group t -test or Welch's t -test were performed (Supplementary Data Table S5). Significance was indicated by $P < 0.05$. Analyses were performed in MATLAB using the `swtest`- (BenSaïda, 2009), `vartestn`- (The MathWorks, Inc, 2018) and `ttest2`-functions (The MathWorks, Inc, 2017), respectively.

RESULTS

We tested a selection of 48 maize genotypes with contrasting above and belowground trait expressions in their water use regulation during soil drying.

The soil used in the present experiment is characterized by a relatively high water-holding capacity, with a saturated water content of 53 vol.-%. The gradual decrease in water potential with decreasing water content is typical for a clay loam (Fig. 5A). The soil hydraulic conductivity curve highlights that below a soil water potential of −10 hPa, the soil hydraulic conductivity drops significantly, i.e. by eight orders of magnitude (Fig. 5B).

As a prerequisite to testing if the drop in belowground hydraulic conductivity during soil drying triggers the downregulation of transpiration, we investigated a selection of above and belowground plant traits that are expected to largely impact the development of water potential gradients

around the roots in drying soils. Our results demonstrate a high genotypic variability within genotypes of all investigated plant traits (Fig. 6). The genotypes comprise a gradient,

where immediately neighbouring genotypes do not differing substantially, but those from the extreme ends of the gradient show considerable contrasts between the median per

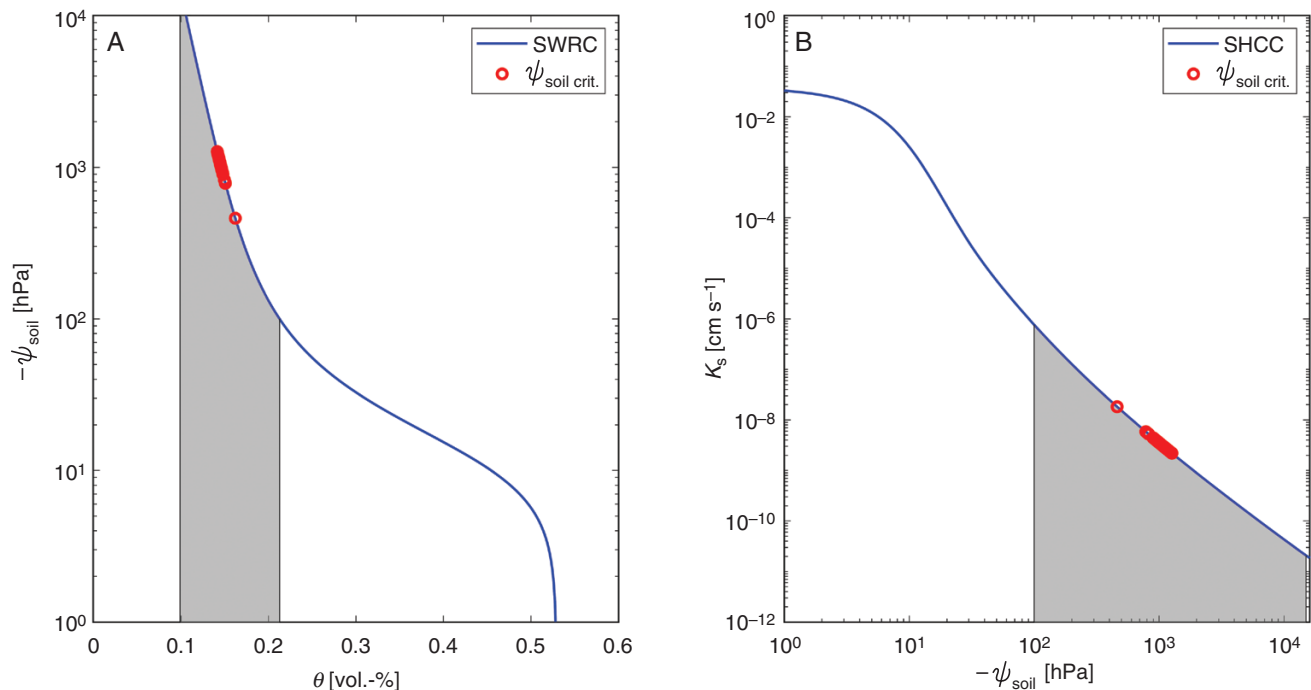


FIG. 5. For the used soil: (A) soil water retention curve (SWRC) – relationship between soil water potential (Ψ_{soil}) and soil water content (θ); and (B) soil hydraulic conductivity curve (SHCC) – relationship between soil hydraulic conductivity (K_s) and soil water potential (Ψ_{soil}). Red circles represent the range of mean $\Psi_{\text{soil crit.}}$ per genotype upon which plants started decreasing transpiration. The grey shaded area marks the plant-available water in the range of Ψ_{soil} between -100 and $-15\,000$ hPa.

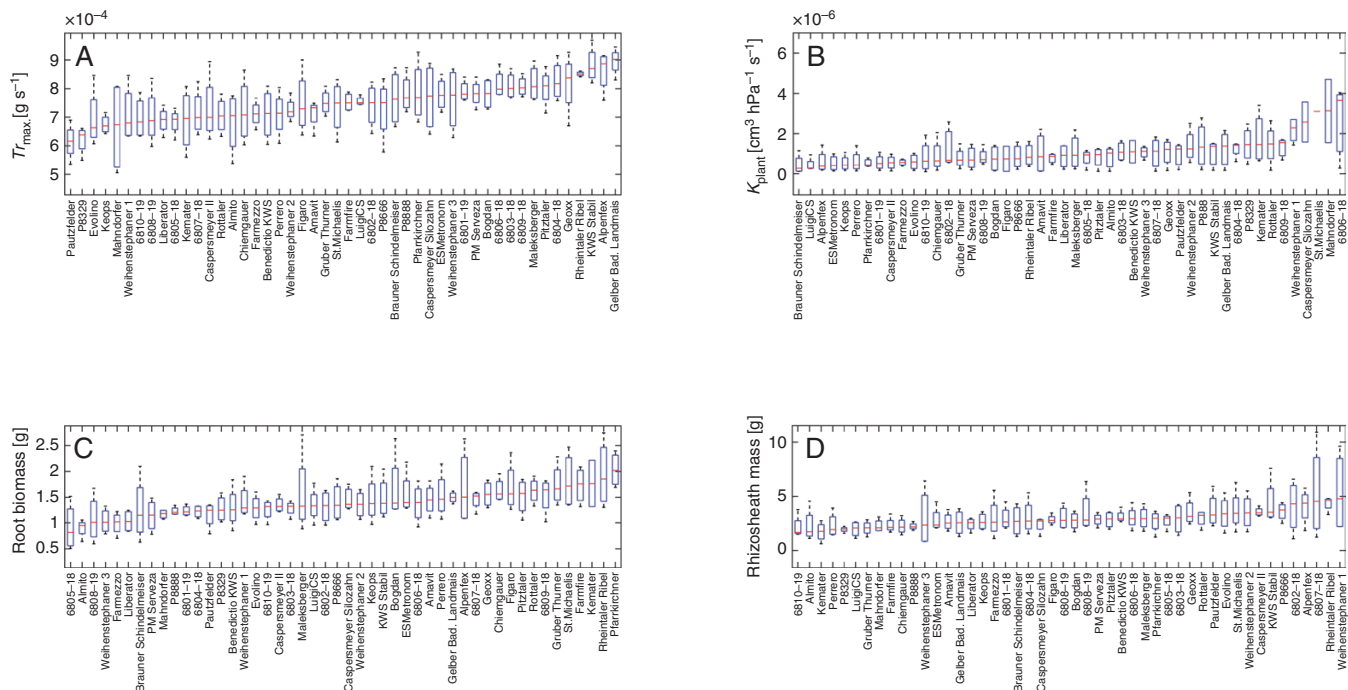


FIG. 6. Above and belowground plant traits that are expected to affect the water potential gradients around the roots during soil drying. Overview of the median and the 25th and 75th percentiles of: (A) maximum transpiration (T_{max}), (B) plant hydraulic conductance (K_{plant}), (C) root dry biomass and (D) rhizosheath dry mass.

genotypes for Tr_{max} (Fig. 6A, min. 6.16×10^{-4} vs. max. 9.02×10^{-4} g s $^{-1}$) and root biomass (Fig. 6C, min. 0.82 vs. max. 2.02 g). For K_{plant} (Fig. 6B, min. 0.28×10^{-6} vs. max. 3.64×10^{-6} cm 3 hPa $^{-1}$ s $^{-1}$) and rhizosheath mass (Fig. 6D, min. 1.66 vs. max. 4.79 g, Supplementary Data Table S5), marginal non-significant trends were visible. Rarely ever were the extreme ends of the gradient of one trait composed of the same genotypes, which makes it hard to draw conclusions in a trivial way (e.g. explaining one response variable with one explanatory variable). Therefore, we aimed to find a mechanistic connection between the selected plant traits and a drought performance indicator.

The critical soil water potential ($\Psi_{soil \text{ crit.}}$) at which plants decreased transpiration upon soil drying was considered to be an indicator of plant water use regulation between genotypes. It ranges from -461 hPa for genotypes that closed stomata earliest in the drying process to -1268 hPa for genotypes that kept stomata open in the driest conditions (Fig. 7). The differences in median $\Psi_{soil \text{ crit.}}$ between genotypes that compose the extreme ends of this gradient are considerable (Supplementary Data Table S5). Plants that closed stomata later during soil drying experienced a soil water potential that was 2.75 times more negative and a soil hydraulic conductivity that was more than one order of magnitude smaller (2.22×10^{-9} vs. 2.52×10^{-8} cm s $^{-1}$) than for plants that decreased transpiration in wetter soil conditions (Fig. 5B). Beyond $\Psi_{soil \text{ crit.}}$, there still was a significant amount of water left to transpire, considering the relatively central position of $\Psi_{soil \text{ crit.}}$ within the theoretical range of plant-available soil water (grey shaded area, Fig. 5A).

$\Psi_{soil \text{ crit.}}$ and the single traits that are hypothesized to impact the drop in belowground hydraulic conductivity during soil drying show the same trend in their relationship to each other (albeit not significant, $P > 0.05$, Supplementary

Data Fig. S4) compared to what we hypothesized (Fig. 1), namely: $\Psi_{soil \text{ crit.}}$ was higher (less negative) if Tr_{max} and K_{plant} were high, and $\Psi_{soil \text{ crit.}}$ was lower (more negative) when the product of root biomass and rhizosheath mass was high. Looking at the relationship between $\Psi_{soil \text{ crit.}}$ and the combination of traits that are hypothesized to impact the development of water potential gradients around roots during soil drying shows that they are considerably positively associated (Fig. 8). Plants with a higher Tr_{max} and K_{plant} and at the same time a smaller root system and rhizosheath mass decrease transpiration at significantly higher (less negative) soil water potentials. Around 43 % of the variability in the data is explained by the equation composing the explanatory variable (Fig. 8, outliers are not considered in the linear fit; Supplementary Data Figure S5).

Finally, Fig. 9 shows the relationship between pre-dawn leaf water potential (Ψ_{PD}) and soil water potential ($\Psi_{soil \text{ meas.}}$) as measured by the soil water potential sensors under pre-dawn conditions. In the wet range (> -1000 hPa), Ψ_{PD} was more negative than $\Psi_{soil \text{ meas.}}$, while the measured soil water potential tends to (absolutely) overestimate the water potential as experienced by the plant (Ψ_{PD}) in the dry range (< -1000 hPa).

DISCUSSION

We investigated the impact of soil hydraulic limitation on plant water use in drying soil. We found that genotypes differed in the critical soil water potential at which plants started decreasing transpiration and that it was related to a combination of above and belowground traits: genotypes with a higher maximum transpiration and plant hydraulic conductance as well as a smaller root system closed stomata at less negative soil water potentials.

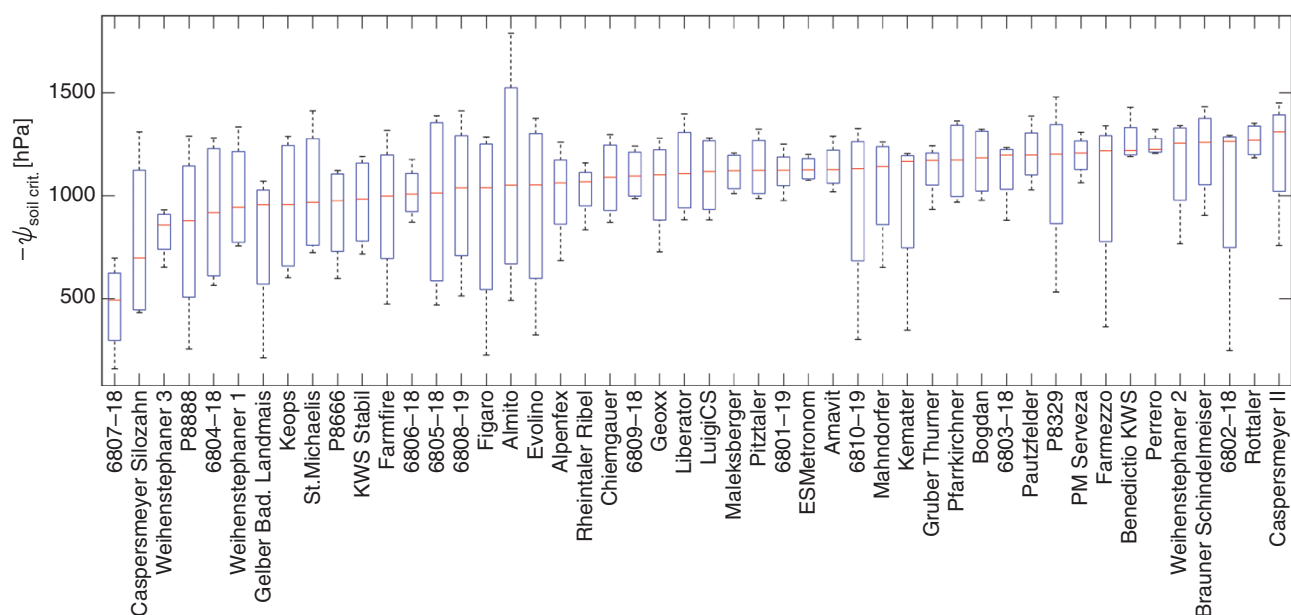


FIG. 7. Maize genotypes compared in their water use regulation in response to drought. The y-axis displays the critical soil water potential ($\Psi_{soil \text{ crit.}}$) at which plants started decreasing transpiration upon soil drying.

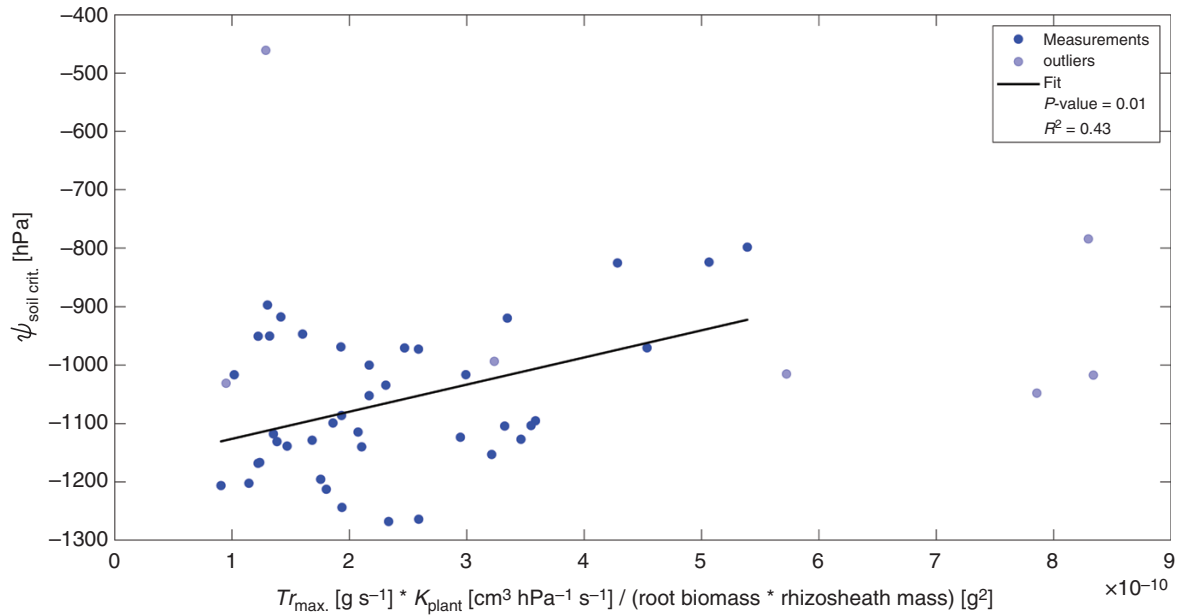


FIG. 8. Relationship between the critical soil water potential at which plants started decreasing transpiration upon soil drying ($\Psi_{\text{soil crit.}}$) and a combination of above and belowground plant traits that are expected to impact water potential gradients around roots during soil drying: maximum transpiration ($Tr_{\text{max.}}$), plant hydraulic conductance (K_{plant}), root dry biomass and rhizosheath dry mass. Each point represents the average over four replicates per genotype. The P -value being < 0.05 indicates the statistical significance of the relationship. The value of R^2 indicates that 43 % of the variability in the data can be explained by the equation composing the explanatory variable. Outliers (Supplementary Data Fig. S5) are not considered in the linear fit. Considering outliers leads to a P -value of 0.06 and R^2 of 0.27.

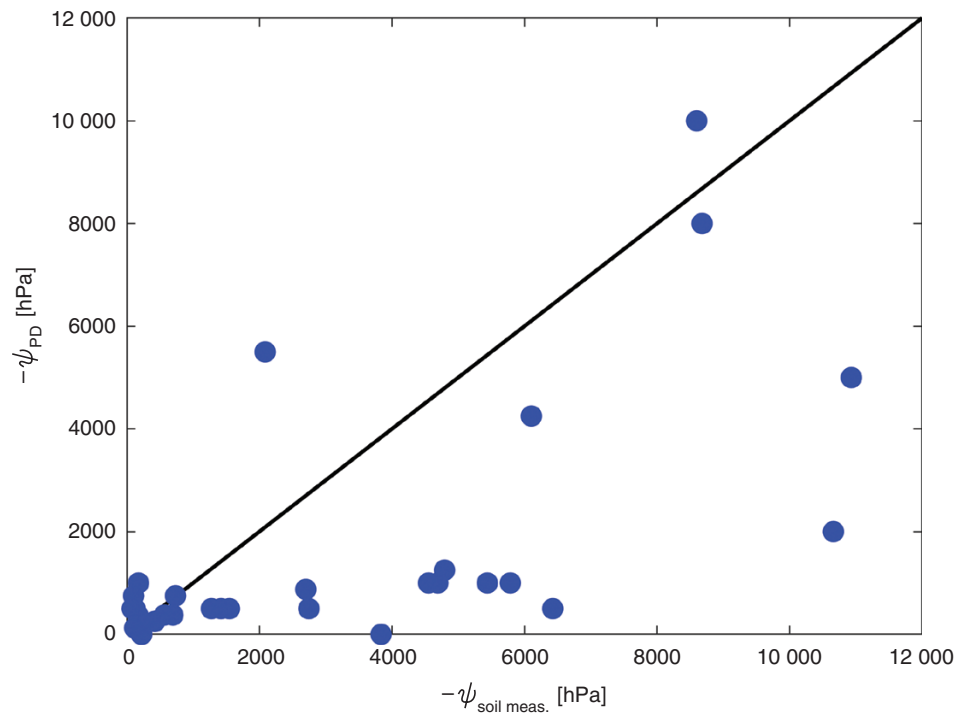


FIG. 9. Relationship between pre-dawn leaf water potential (Ψ_{PD}) and soil water potential ($\Psi_{\text{soil meas.}}$) as measured by the soil water potential sensors.

Plant water use regulation during soil drying between genotypes

The critical soil water potential at which plants decreased transpiration varied highly across the genotypes between -461 and

-1268 hPa (Fig. 7). We used the critical soil water potential as an indicator for plant water use regulation and therefore drought responsiveness (Cai *et al.*, 2022). Considering that all used genotypes are from Germany, Austria or Switzerland and have

therefore experienced comparatively small environmental differences in an evolutionary/breeding context, it is not surprising that the differences in expressions of traits between genotypes are rather subtle (Fig. 6). However, the extreme ends of the gradient that the 48 screened genotypes comprise are considerably different in critical soil water potential (Supplementary Data Table S5). This suggests that plants experienced significant differences in water potential around the roots as well as in soil/rhizosphere hydraulic conductivities (Fig. 5B) when they closed stomata. Plants differed considerably in the time until they showed symptoms of drought stress and responded to drought. To shed light on the physical constraints to water flow during soil drying, we investigated the relationship of drought responsiveness and plant traits that impact belowground hydraulic conductivity and development of water potential gradients around roots during soil drying.

Plant above and belowground traits affecting transpiration response to soil drying

We found that the critical soil water potential was higher (meaning that plants closed stomata comparatively earlier/in wetter soil conditions during soil drying) for plants that exhibited a higher maximum transpiration and plant hydraulic conductance while at the same time having a smaller root mass and rhizosheath mass (Fig. 8). The ratio between maximum transpiration and the combination of root mass and rhizosheath mass represents a typical demand and supply scheme.

Maximum transpiration defines plant water demand. Under comparable atmospheric conditions, maximum transpiration under optimal soil water availability between genotypes depends on leaf and stomatal properties such as leaf area and stomatal density and size (Devi and Reddy, 2018; Tardieu, 2022). In drying soils, maximum transpiration impacts the belowground hydraulic conductivity around roots by being the driving force of water flow. Thereby, it dictates the water flow velocity, and the development of water potential gradients at the soil–root interface.

Root surface area that is effective in water uptake determines plant water supply. A large root surface area attenuates water potential gradients in the rhizosphere due to a lower water flux at the root–soil interface and thereby softens the drop in belowground hydraulic conductivity (Abdalla et al., 2022; Cai et al., 2022). However, it was shown that rather than root system size alone, root hydraulics and the fraction of roots effectively involved in water uptake determine the overall water uptake of crops (Vadez, 2014; Ahmed et al., 2018). The fraction of roots effectively contributing to water uptake is hardly quantifiable as roots do not take up water uniformly (Ahmed et al., 2018). Indeed, for maize it has been shown by means of simulations that the fraction of root length effectively taking up water ranges merely between 1 and 2% (Hayat et al., 2019; Cai et al., 2021). Root biomass, indicating root length (Supplementary Data Fig. S3), is a good indicator for soil exploration on the field scale (Lynch, 2013) and a parameter that is relatively easy to measure in a large belowground phenotyping experiment, while effective root length can only be obtained from modelling (Ahmed et al., 2018; Abdalla and Ahmed, 2021; Cai et al., 2021) and root hydraulic conductivity is usually measured under controlled conditions on a much smaller scale (Tsuda and Tyree,

2000; Rodriguez-Dominguez and Brodribb, 2020; Abdalla et al., 2022). Additionally, processes in the rhizosphere might extend the effective root radius and thereby attenuate the drop in water potential around the roots with soil drying (Carminati et al., 2017). Therefore, we decided to use a combination of root biomass dry weight and rhizosheath mass as an approximation for the root surface area effective in water uptake.

Rhizosheath is defined as the soil that adheres to the roots after extracting the roots from the soil and depends on properties of the roots and the soil in the immediate vicinity of the roots, such as root hair length and density, rhizobacterial exopolysaccharide production, root exudation, and mycorrhizal fungi (Ndour et al., 2020; Aslam et al., 2022). It was shown that exudates in the rhizosphere increase the unsaturated hydraulic conductivity under drying conditions and thereby favour root water uptake in relatively dry soils (Ahmed et al., 2014). Furthermore, root hairs have been proposed to be beneficial during soil drying (Carminati et al., 2017; Marin et al., 2021) by enhancing the soil–root contact (Duddek et al., 2022), just as hyphae of mycorrhizal fungi can be seen as an extension of the root system (Vidal et al., 2018; Augé, 2001; Abdalla et al., 2022). The strength of rhizosheath mass expression as a combination of the many rhizosphere processes is expected to impact the steepness of the drop in rhizosphere hydraulic conductivity. In summary, the lower root mass and rhizosheath mass, the more limited is the water supply to the plant, and stomata will need to close in wetter soil conditions.

Intuitively, it makes sense that the ratio of supply and demand will impact plant water use regulation. If the demand is very high, but the supply is limited by a small effective root system, the plant experiences more severe gradients of water potential at similar soil water contents and will close stomata at comparatively higher water contents already. If the plant has sufficient access to water, and a limited maximum capacity to transpire, it can sustain transpiration even under lower soil water contents. Additionally, we found that plants decrease transpiration at higher soil water potentials (meaning in wetter soil conditions) if their plant hydraulic conductance is high. This seems counterintuitive.

Plant hydraulic conductance is a key trait to evaluate plant water use regulation (Cai et al., 2022) and includes the conductance of aboveground xylem, and roots. It is shaped by axial (dictated by xylem vessel size and abundance; Richards and Passioura, 1989) and radial conductance (Doussan et al., 2006), dictated by, for example, aquaporin expression (McLean et al., 2011), the formation of Casparian bands (Zimmermann et al., 2000) and suberization (Kreszies et al., 2019), as well as root architecture. Considering the theory of Sperry and Love (2015), Sperry et al. (2016) and Carminati and Javaux (2020), stomatal regulation is proposed to prevent plants from experiencing a non-linear drop in the supply function, as triggered by a loss in the soil–plant hydraulic conductance. Applying this concept, plants with a higher hydraulic conductance are expected to show a decrease in transpiration at higher water potentials during soil drying, which was also reported by Choudhary and Sinclair (2014) and Cai et al. (2022), for example. Considering the SPAC as a system of hydraulic resistances in series, for which the overall resistance is equal to the sum of the single resistances of the compartments of the SPAC, plant hydraulic resistance (i.e. the inverse of the hydraulic conductance) will

determine how susceptible plant transpiration is to a change in total conductance. For plants with low hydraulic resistance, soil hydraulic resistance will more quickly become the limiting factor for the overall resistance in such a system in series. Plants with a high conductance are therefore expected to be more sensitive to an increase in water potential gradients around roots in drying soils (i.e. the decline in soil hydraulic conductivity). This suggests that a plant with a higher hydraulic conductance is more susceptible to a change in total conductivity, which is a logical consequence of the hypothesis that plants close stomata in response to a drop in conductivity (Fig. 1B). However, this theory holds true only for low VPD levels and dry soil conditions, when the largest dissipation of water potentials along the soil–plant continuum occurs in the soil, rather than in the plants. Applying the same concept in well-watered conditions at high VPD, when the dissipation of water potential is mostly taking place within the plant, a low plant (or root) hydraulic conductance will cause a very negative leaf water potential, which might be sufficient to induce non-linearities in the leaf hydraulic conductance and trigger stomatal closure (Richards and Passioura, 1989). The counterintuitive result that plants with higher hydraulic conductance reduce transpiration earlier than plants with low conductance is therefore true only for low VPD conditions. For high VPD in wet soils, the opposite relationship is expected.

Link between plant water use and above and belowground hydraulics during soil drying

The relationship between the critical soil water potential and the single traits was not statistically significant, albeit showing the expected trends (Supplementary Data Fig. S4). This might be due to the single traits overlapping each other in their effects on the critical soil water potential. When combining the traits based on the apparent trends and based on our hypothesis, the relationship between the critical soil water potential and the combination of the investigated traits was statistically significant and explained 43 % of the variability in the data (Fig. 8). The unexplained rest of the variability might be related to: (1) uncertainties in the indirect determination of soil water potential from soil water content. On the one hand, this is due to the non-linear nature of the soil water retention curve, especially in dry conditions (Novick *et al.*, 2022). On the other hand, the uncertainties in the exact determination of soil water potential might be due to potential marginal heterogeneities in water distribution within the mesocosms (Cai *et al.*, 2021). The unexplained rest of the variability might moreover be related to: (2) uncertainties due to slight differences in water content of the rhizosheath during sampling (On the other hand, the uncertainties in the exact determination of soil). More importantly, the additional scatter in the data might be related to (3) uncertainties regarding the estimation of root surface area effective in water uptake from total root biomass. It cannot be expected that the entire root system was effective in water uptake and the fraction of roots effectively taking up water might differ between genotypes, as has been shown for maize (Ahmed *et al.*, 2016, 2018). Lastly, the unexplained rest of the variability might be related to (4) uncertainties regarding plant traits that have not been

investigated, such as stomatal properties (e.g. density and size) or active mechanisms that are known to impact stomatal closure (e.g. hormonal signalling; Brodribb and McAdam (2017)). However, having found a significant relationship between the critical soil water potential and the combination of maximum transpiration, plant hydraulic conductance, root mass and rhizosheath mass seems to support the link between plant water use during soil drying and belowground hydraulic conductivity. This becomes especially obvious considering the multiple potential sources of uncertainties that come with a large-scale phenotyping experiment.

Offset between pre-dawn leaf water potential and soil water potential

Finally, we found a rather large offset between pre-dawn leaf water potential and the measured soil water potential. This deviation between pre-dawn leaf water potential and measured soil matric potential has often been reported (Carminati *et al.*, 2017; Cai *et al.*, 2020; Abdalla and Ahmed, 2021; Koehler *et al.*, 2022). In the wet range, the pre-dawn leaf water potential was slightly more negative than the measured soil water potential (Fig. 9). This might be related to non-negligible night-time transpiration (as demonstrated for maize, e.g. by Tamang and Sadok, 2018) potentially having induced minor fluxes and therefore more negative pre-dawn leaf water potentials. Additionally, the determination of pre-dawn leaf water potential in wet soil conditions involves uncertainties introduced by the low resolution of the Scholander Bomb below one bar (1000 hPa). Moreover, the soil matric potential sensors (TEROS 21) are not accurate in the wet range, which might add to the offset. The soil water potential sensors solely measure the soil matric potential, thereby neglecting the osmotic component of the total soil water potential. Therefore, the offset between the pre-dawn leaf water potential and the soil water potential could reflect the osmotic potential of the soil solution (Carminati *et al.*, 2017). The pre-dawn leaf water potential having been more negative than the soil water potential would suggest that the osmotic potential was less negative in the soil–root zone than in the xylem, which is well in line with observed guttation on the well-watered plants under pre-dawn conditions, as in Singh (2016). In the drier range, the measured soil water potential was considerably more negative than the pre-dawn leaf water potential as the soil was drying (Fig. 9). Although it is generally accepted to assume one soil water potential for a whole pot due to its limited extent, the soil water might have been distributed slightly heterogeneously (Cai *et al.*, 2021). If the soil water potential sensor was installed in a slightly drier region, it would have read lower numbers, which would result in an underestimation of what the plant has experienced on average. The pre-dawn leaf water potential can serve as an integrative measure of the total water potential as sensed by the plant, also in the case of a heterogeneous soil moisture distribution (Couvreur *et al.*, 2012). Based on these considerations, we used pre-dawn leaf water potential rather than the measured soil water potential for the inference of plant hydraulic conductance (if using soil water potential to infer plant hydraulic conductance; see the result in Supplementary Data Fig. S7).

CONCLUSIONS

This study underlines the importance of the drop in soil hydraulic conductivity as impacted by different above and belowground traits for plant water use regulation in response to soil drying. We found that the tested maize genotypes decreased transpiration at different soil water potentials and identified a combination of plant and rhizosphere traits that impact the physical constraints to water flow through soils and plants during soil drying, namely: maximum transpiration, plant hydraulic conductance, and root mass and rhizosheath mass. Plants decreased transpiration at higher soil water potentials when they exhibited a relatively higher maximum transpiration and plant hydraulic conductance and at the same time were characterized by a relatively small root system effective in water uptake (e.g. low root biomass and rhizosheath mass).

According to the concept of Carminati and Javaux (2020) and Sperry and Love (2015), the critical soil water potential corresponds to the point at which plants downregulate transpiration. This implies that plants with a relatively higher conductance will be more responsive to soil drying and will hence decrease transpiration at comparatively higher (less negative) soil water potentials, at least for low VPD. The results are in line with our hypothesis. Our analyses improve our understanding of the mechanisms and below and aboveground hydraulic traits that control transpiration response to soil drying. Taken together, this study highlights the role of belowground hydraulic conductivity as well as related above and belowground plant traits and their interactions in crop water use. Ultimately, the effect of plant traits on drought tolerance also depends on environmental conditions and agricultural management practices.

SUPPLEMENTARY DATA

Supplementary data are available online at <https://academic.oup.com/aob> and consist of the following.

Note S1: quantification of biomass dry weights.

Note S2: quantification of rhizosheath dry weight.

Figure S1: relationship between transpiration and soil water content for all 48 tested genotypes, including the range of calculated critical soil water contents and confidence intervals of the sigmoidal fit and θ_{crit} , respectively.

Figure S2: images of the intact soil block after taking it out of the mesocosm; carefully extracted root system hanging in the 'shakinator' that shook them at a defined intensity and time; roots after shaking with rhizosheath remaining.

Figure S3: relationship between root biomass and root length as obtained from the same experiment for 38 out of the 48 genotypes and from the opposite soil compartment of the same mesocosm.

Figure S4: relationship between the critical soil water potential and the investigated above and belowground traits that are expected to impact transpiration behaviour during soil drying: maximum transpiration rate, plant hydraulic conductance, and the product of root dry biomass and rhizosheath dry mass.

Figure S5: overview of the median and the 25th and 75th percentiles of the critical soil water potential upon which plants decreased their transpiration, maximum transpiration, plant

hydraulic conductance, and the product of root biomass and rhizosheath mass.

Figure S6: histogram of the soil water content as measured at the beginning of sampling of the maize phenotyping experiment.

Figure S7: relationship between the critical soil water potential at which plants started decreasing transpiration upon soil drying and a combination of above and belowground plant traits that are expected to impact water potential gradients around roots during soil drying: maximum transpiration, plant hydraulic conductance, root dry biomass and rhizosheath dry mass.

Table S1: overview of the tested maize varieties.

Table S2: textural properties of the initial soil material as obtained by wet sieving and SediGraph particle size analysis for the soil material sampled at mesocosm filling and for the soil material sampled directly after excavation.

Table S3: overview of parameters as fitted using the Peters Durner Iden parameterization for characterizing the soil hydraulic properties.

Table S4: chemical properties of the initial soil material: soil organic carbon, nitrogen concentration, carbon-to-nitrogen ratio and $\delta^{13}C$.

Table S5: outcome of statistical testing for significance between extreme genotypes in their expression of evaluated above and belowground traits: critical soil water content and potential at which plants decreased transpiration, plant hydraulic conductance, maximum transpiration, root biomass and rhizosheath mass.

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CONFLICT OF INTEREST

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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